Sri Chaitanya IIT Academy.,India.

A.P T.S KARNATAKA TAMILNADU MAHARASTRA DELHI RANCHI

A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

SEC: Sr.S60(N&S), Elite, IAS & LIIT | JEE-MAIN-Shift-1 | Date: 26-03-2026

Time: 09.00Am to 12.00Pm | GTM-29 | Max. Marks: 300

KEY SHEET

MATHEMATICS

1	**2**	2	**1**	3	**2**	4	**4**	5	**3**
6	**1**	7	**4**	8	**3**	9	**2**	10	**4**
11	**2**	12	**3**	13	**3**	14	**1**	15	**2**
16	**3**	17	**1**	18	**3**	19	**4**	20	**1**
21	**108**	22	**10**	23	**5**	24	**7**	25	**3**

PHYSICS

26	**1**	27	**3**	28	**3**	29	**1**	30	**2**
31	**1**	32	**4**	33	**3**	34	**1**	35	**2**
36	**4**	37	**3**	38	**2**	39	**1**	40	**3**
41	**1**	42	**4**	43	**3**	44	**2**	45	**1**
46	**51**	47	**7**	48	**6**	49	**7**	50	**4**

CHEMISTRY

51	**1**	52	**2**	53	**1**	54	**3**	55	**2**
56	**1**	57	**4**	58	**2**	59	**2**	60	**4**
61	**4**	62	**3**	63	**3**	64	**4**	65	**1**
66	**2**	67	**1**	68	**3**	69	**3**	70	**1**
71	**2**	72	**4**	73	**1**	74	**55**	75	**138**

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SOLUTIONS

MATHEMATICS

1. $[x]+\left[\frac{x}{2}\right]=4,\left[\frac{x}{3}\right]=1\Rightarrow x\in[3,4)$

   $a=3,b=4\Rightarrow ab=12$

2. $\log y=\frac{3n^3+4}{4n^4-1}\sum_{r=1}^{n}\log\left(\frac{r}{n}\right)=\frac{3}{4}\int_0^1 \ell nx\,dx=\frac{-3}{4}\Rightarrow y=e^{-3/4}$

3. Are $a=4\int_0^1\sqrt{x^2-x^6}\,dx=4\int_0^1 x\sqrt{1-x^4}\,dx$ $put\ x^2=\sin\theta$

   Area $=\frac{\pi}{2}$

4. $2^{10}6^{20}=1024(7-1)^{20}=1024(7k+1)=7k^1+1022+2=7m+2$

5. (A) possible non –negative values if |A| = 2,4,8

   (B) Sum is 0

   (C) |adj (adj (adj A))|=|A| $\Rightarrow$least value |A| =2

   Possible values are ± 2

   (D) $\frac{16}{|A|}=\frac{16}{-8}=-2$

6. Statement -1 : 33 x 1111 =36663

   Statement -2 : $={}^9C_2\times{}^9C_2-\sum 8^2=1092$

7. $2^x=3^y=6^{-z}=k$

   $x=\log_2 k, y=\log_3 k, z=-\log_6 k$

   $\Rightarrow\frac{1}{x}+\frac{1}{y}+\frac{1}{z}=0$

8. $y=9+2\left(\tan^2 x+\cot^2 x\right)$

   $y\geq 9+2(2)$

   $y\geq 13$

9. $y=3\sin^{-1}x, y=\pi(1-x)$, Graphically no. of solutions is 1

10. $1-\frac{1}{2}\times\frac{2}{3}\times\frac{3}{4}=1-\frac{1}{4}=\frac{3}{4}$

11. $x_1x_2x_3=2^3.3^1.5^1={}^{3+3-1}C_{3-1}\times{}^{1+3-1}C_{3-1}\times{}^{1+3-1}C_{3-1}={}^5C_2\times{}^3C_2\times{}^3C_2=10\times3\times3=90$

12. $\lim_{x\to2}\frac{(x-2)(ax+b)}{\sin(x-2)}=7$ =2a + b = 7 =3a + b = 9$\Rightarrow a=2,b=3\Rightarrow P(x)=(x-2)(2x+3)$

    $P(4)=2\times11=22$

13. $2x=2\frac{dy}{dx}=\frac{dy}{dx}=x$ =m = 1= x = 1= $y=\frac{1}{2}=P\left(1,\frac{1}{2}\right)$

    Perpendicular distance of $\left(1,\frac{1}{2}\right)$ *from* $x-y-1=0$ is

    $\frac{1}{2\sqrt{2}}\simeq 0.3535$

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14. $\lambda = 3 \;=\; \bar{v}_1 = k\bar{u} \;=\; \bar{v}_2.\bar{u} = 0 \qquad \left(\bar{v} - \bar{v}_1\right).\bar{u} = 0 = \qquad \bar{v}.\bar{u} = \bar{v}_1.\bar{u}$

$\bar{v}.\bar{u} = k.\left|\bar{u}\right|^2 \qquad 5 = \mathrm{k}(10) \qquad k = \frac{1}{2} \;=\; \bar{v}_1 = \frac{1}{2}\bar{u} \;=\; \bar{v} = \frac{\bar{u}}{2} + \bar{v}_2 \;=\; \bar{v}_2 = \frac{2\bar{v} - \bar{u}}{2}$

$\left|\bar{v}_1\right|^2 + \left|\bar{v}_2\right|^2 = 14 \qquad k = 14 = 2^1 \times 7^1$

no. of divisions = 2 x 2=4

15. $\left|\lambda - 3\right| = 1 \qquad \lambda_1 = 4, \lambda_2 = 2$

Let Equation of circle passing through (0, 0) , (4,2), (2,4) be

$x^2 + y^2 + 29x + 2fy + c = 0 \Rightarrow c = 0, 9 = f = \frac{-5}{3}$

$r = \sqrt{\frac{25}{9} \times 2} = \sqrt{\frac{50}{9}} \qquad \therefore Area = \frac{50\pi}{9}$

16. $3f(x) = xf'(x) + f(x) - x^2$

$xf'(x) - 2f(x) = x^2 \quad f'(x) - \frac{2}{x}f(x) = x$

$I.F = e^{\int \frac{-2}{x}dx} = \frac{1}{x^2} \qquad y\left(\frac{1}{x^2}\right) = \int x.\frac{1}{x^2}dx + c$

$\frac{y}{x^2} = \log x + c \quad y = x^2 \log x + cx^2$

$f(1) = \frac{1}{3} \Rightarrow c = \frac{1}{3} \qquad y = x^2 \log x + \frac{x^2}{3} \qquad y(e) = \frac{4}{3}e^2$

17. $I = \int_0^{1/2} \frac{\left(1+\sqrt{3}\right)dx}{(1+x)^{1/2}(1-x)^{3/2}} \qquad I = \int_0^{1/2} \frac{\left(1+\sqrt{3}\right)dx}{(1+x)^2\left(\frac{1-x}{1+x}\right)^{3/2}} \qquad \frac{1-x}{1+x} = t \quad \frac{-2dx}{(1+x)^2} = dt$

$I = \left(1+\sqrt{3}\right)\int_1^{1/3} \frac{dt}{-2t^{3/2}} = -\frac{\left(1+\sqrt{3}\right)}{2}\left(\frac{-2}{\sqrt{t}}\right)\Bigg]_1^{1/3} = 2$

18. $r = \frac{r_1 r_2}{\left(\sqrt{r_1} + \sqrt{r_2}\right)^2} = \frac{36 \times 9}{(6+3)^2} = \frac{36 \times 9}{9x9} = 4$

19. $y_1 = 4e^{4x} - 2e^{-x} \qquad y_2 = 16e^{4x} + 2e^{-x} \qquad y_3 = 64e^{4x} - 2e^{-x}$

$y_3 - 13y_1 = 64e^{4x} - 2e^{-x} - 13\left(4e^{4x} - 2e^{-x}\right)$

$y_3 - 13y_1 = 64e^{4x} - 2e^{-x} - 13\left(4e^{4x} - 2e^{-x}\right) = 64e^{4x} - 2e^{-x} - 52e^{44} + 26e^{-x} = 12e^{4x} + 24e^{-x}$

$\frac{y_3 - 12y_1}{2y} = \frac{12\left(e^{4x} + 2e^{-x}\right)}{2\left(e^{4x} + 2e^{-x}\right)} = 6$

Tan x = 6 $\Rightarrow$ Graphically no of solutions = 2

20. $\left(e^x\right)^3 - 11\left(e^x\right)^2 + \frac{81}{2}e^x - 45 = 0 \qquad e^x = y$

$y^3 - 11y^2 + \frac{81}{2}y - 45 = 0 \qquad y_1 y_2 y_3 = 45$

$e^{x_1}.e^{x_2}.e^{x_3} = 45 \;\; x_1 + x_2 + x_3 = \log 45$

Log P = log 45 $\Rightarrow P = 45$

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21. $\sin\theta = \frac{d}{3}$ $\quad \ell = 4a\cos ec^2\theta$

$\ell = 12 \times \frac{9}{d^2}$ $\quad \ell d^2 = 108$

22. $1 + \frac{6z}{2 - 3z + 4z^2} \Rightarrow \frac{z}{2 + 3z + 42^2}$ must be real

$\Rightarrow \frac{z}{2 - 3z + 4z^2} = \frac{\bar{z}}{2 - 3\bar{z} + 4\bar{z}} \Rightarrow |z|^2 = \frac{1}{2} \Rightarrow 20|t|^2 = 20 \times \frac{1}{2} = 10$

23. Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$

a + d =3

det (A) =ad – bc

$tr(A^3) = a^3 + abc + bca + bcd + bca + bcd + a^3 + bcd = (a+d)^3 - 3(a+d)(ad - bc)$

$-18 = 3^3 - 3.3\det(A)$

$9\det(A) = 27 + 18 = 45 = \det(A) = 5$

24. $b = ar, c = ar^2$

$\frac{a+b+c}{3} = b + 2$

$a + ar + ar^2 = 3ar + 6$

$ar^2 - 2ar + a = 6$

$r^2 - 2r + 1 = \frac{6}{a}$

$(r-1)^2 = \frac{6}{a}\ a \in N$

a = 6

$(r-1)^2 = 1$

$r - 1 = z1$

$r = 2$

$\Rightarrow a + 1 = 6 + 1 = 7$

25. $xd3 - ydx = y^2dy$

$\frac{ydx - xdy}{y^2} = -dy$

$d\left(\frac{x}{y}\right) = -dy$

$\frac{x}{y} = -y + c$

$y(1) = 1 \Rightarrow c = 2\ \frac{x}{y} + y = 2$

Put x = -3 $= \frac{3}{y} + y = 2$

$y^2 - 2y - 3 = 0$

$(y-3)(y+1) = 0$

y = 3

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PHYSICS

26. The given inductors are in parallel in the circuit, and so

$L_{eq} = \frac{L}{3} = \frac{3}{3} = 1H.$

27. Force on $P = -\frac{kQqz_0}{\left(R^2 + z_0^2\right)^{3/2}}$

Since directed towards origin always it is periodic

For $z << R, F \simeq \frac{-kQq}{R^3} z_0$

Hence S.H.M

28. If L be the length of the lake and the velocity of boat is V, time taken in going and coming back on a quiet day

$t_Q = \frac{L}{V} + \frac{L}{V} = \frac{2L}{V} \ \ldots (1)$

Now, if υ is the velocity of air-current, then time taken in going across the lake

$t_1 = \frac{L}{V + \upsilon}$ (as the current helps the motion) and time taken in coming back

$t_2 = \frac{L}{V - \upsilon}$

So $t_R = t_1 + t_2 = \frac{2LV}{V^2 - \upsilon^2} = \frac{2L}{V\left[1 - \left(\frac{\upsilon}{V}\right)^2\right]} \ldots (2)$

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Hence from equations (1) and (2),

$\frac{t_R}{t_Q} = \frac{1}{1 - (\upsilon/V)^2} > 1 \left[\because 1 - \left(\frac{\upsilon}{V}\right) < 1\right]$

$\therefore t_R > t_Q$

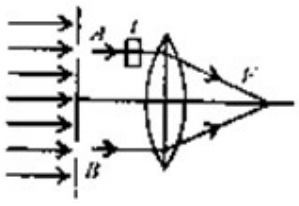
29. A and B are two circular apertures. A film of thickness t is placed in front of aperture A. F is the focal spot of the lens. To calculate power received at the focal point, we need to determine the power transmitted by the apertures. Let power received by $A = P_A$

$\therefore P_A = I \times \pi r^2{}_A$ or $P_A = \left(\frac{10}{\pi}\right)(\pi)(0.001)^2$ or $P_A = 10^{-5} W$

$\therefore$ power transmitted by $A = \frac{10}{100} \times 10^{-5}$ or $P_1 = 10^{-6} W$

Similarly $P_B = I \times \pi (r_B)^2$ or $P_B = \left(\frac{10}{\pi}\right)(\pi)(0.002)^2 = 4 \times 10^{-5} W$

$P_2 = \frac{10}{100} \times 4 \times 10^{-5} = 4 \times 10^{-6} W$ or $P_2 = 4 \times 10^{-6} W$



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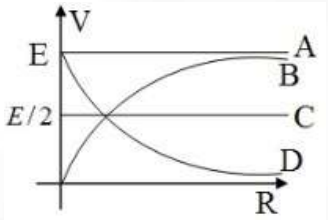
30. $V_P - V_Q = 2E$ for length l, $P.D. \to 2E$

For length $\frac{l}{n}$, $P.D. \to \frac{2E}{n}$

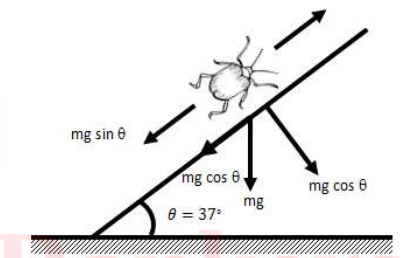
$V_S = V_P - \frac{2E}{n}; V_A = V_P - E$ resistance $\to \frac{R_0}{n}$

$V_S - V_A = E\left(1 - \frac{2}{n}\right)$

$\therefore I_{am} = \left(1 - \frac{2}{n}\right)\frac{E}{r + \frac{R_0}{n}}$

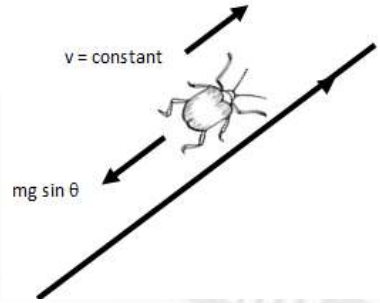


31. $mg\sin\theta + mg\cos\theta = ma$



It can apply maximum equal to frictional force.

$\therefore Fuet = 0$, $m\frac{dv}{dt} = 0$



$V =$ Contact

$a_d = g\sin\theta + g\cos\theta = 10\left[\frac{3}{5} + \frac{4}{5}\right] = 14m/\sec^2$; $a_u = g\cos\theta g\sin\theta = 10\left(\frac{4}{5} - \frac{1}{5}\right) = 2m/\sec^2$

If $f_r = mg\sin\theta = mg \times \frac{3}{5} = \frac{3mg}{5} < f_{r\max}$

$f_r < f_{r\max}$

$\frac{3mg}{5} < \frac{4mg}{5}$

Hence insect can move with constant velocity.

32. Since, $eV_s = hv - W \Rightarrow 5eV = hv - 2eV \Rightarrow hv = 7eV$

So, total number of photons incident is $n = \frac{P}{hv}$

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Since, $\eta = 10^{-3}\% \Rightarrow n_{\text{emitted}} = n \times \frac{10^{-3}}{100} \Rightarrow i = \frac{q}{t} = \left(\frac{n_{\text{emitted}}}{t}\right) e$

Where, from the graph

$$i = 10 \times 10^{-6}\ A \Rightarrow 10 \times 10^{-6} = \frac{P}{hv} \times \frac{10^{-3}}{100} \times 1.6 \times 10^{-19} \Rightarrow 10 \times 10^{-6} = \frac{P \times 10^{-3} \times 1.6 \times 10^{-19}}{7 \times 1.6 \times 10^{-19} \times 100} \Rightarrow P = 7\text{W}$$

33. $\omega = \omega_0 - k\phi$ or $\frac{d\phi}{dt} = \omega_0 - k\phi$

or $\int_0^{\phi} \frac{d\phi}{\omega_0 - k\phi} = \int_0^t dt$

or $-\frac{1}{k}\left[\ln\left(\omega_0 - k\phi\right)\right]_0^{\phi} = t$

or $1n\left(\frac{\omega_0 - k\phi}{\omega_0}\right) = -kt$

$\therefore\ \phi = \frac{\omega_0}{k}\left(1 - e^{-kt}\right)$

34. Let the length of rope to left of pulley initially and finally be l_1 and l_2 as shown

$\therefore\ l_1 = 2\,\text{m}$

and $l_2 = 2\cos 30^\circ + l_1 \cos 30^\circ = 2\sqrt{3}\,\text{m}$

Hence, increase in length of rope $= l_2 - l_1 = 2\left(\sqrt{3} - 1\right)\,\text{m}$

$\therefore$ Block rises up by $2\left(\sqrt{3} - 1\right)\,\text{m}$

Hence, work done by man on rope is

$mg\left(l_2 - l_1\right) = 25 \times 10 \times 2\left(\sqrt{3} - 1\right)$

Therefore work done by rope on man $= 500\left(\sqrt{3} - 1\right)\,\text{J}$

35. When the stone is suspended in air

$$n = \frac{1}{2L}\sqrt{\frac{W_a}{m}}$$

When the stone is suspended in water

$$n = \frac{1}{2L'}\sqrt{\frac{W_w}{m}}$$

Hence, $\frac{\sqrt{W_a}}{L} = \frac{\sqrt{W_w}}{L'}$

or $\frac{W_a}{W_w} = \frac{L^2}{L'^2}$

Now, specific gravity of material of the stone

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$$= \frac{W_a}{W_a - W_w} = \frac{1}{1 - \frac{W_w}{W_a}} = \frac{1}{1 - \frac{L'^2}{L^2}} = \frac{L^2}{L^2 - L'^2} = \frac{(40)^2}{(40)^2 - (22)^2}$$

36. A → r, s, f ; B → q, r, s, t;

C → p,q; D → q, r, s, t

(p)$X_L = 0, V_1 = 0, V_2 = V$

(q)$X_L = 0, V_1 = 0, V_2 = V$

(r)$X_L = \omega L = 2\pi f L = 1.884\Omega, R = 2\Omega$

$V_1 = IX_L$ and $V_2 = IR \quad \therefore \quad V_2 > V_1$

(s)$X_L = 1.884\Omega$

$X_C = \frac{1}{\omega C} = \frac{10}{3\pi} \times 10^3\Omega$

$x_c >> X_L$

$V_2 = IX_C, V_1 = IX_L \Rightarrow V_2 > V_1$

(t)$R = 1\text{k}\Omega \quad X_C = \frac{10}{3\pi} \times 10^3\Omega$

$X_C > R \Rightarrow V_2 > V_1$

Also $V_2 = IX_C, V_1 = IR$

37. Energy is equally divided between electric and magnetic fields of an electromagnetsic wave.

38. Reverse saturation current is due to minority charge carriers. When temperature rises by 100°C, the semiconductor loses the property of controlled conduction due to the breaking of many covalent bonds.Directions : In the following questions (14-15) a statement of assertion is followed by a statement of reason. Mark the correct choice as

(a)If both assertion and reason are true and reason is the correct explanation of the assertion.

(b)If both assertion and reason are true but reason is not correct explanation of the assertion.

(c)If assertion is true, but reason is false.

(d)If both assertion and reason are fa IRV assertion is false and reason is true.

39. Both the assertion and reason are true and reason is the correct explanation of the assertion.

40. According to Wien's displacement law,

$\lambda_m \propto \frac{1}{T}$

The assertion true but the reason is false.

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41. X-rays' frequency is more than that of UV rays. So, both K_{max} and stopping potential increase. Statement-2 is incorrect. Photoelectrons are emitted with a range of kinetic energies because different electrons have different surface binding energies.

42. $K_{max} = hv - hv_0$

When frequency is doubled,

$K'_{max} = h\times 2v - hv_0 = hv + (hv - hv_0) = hv + K$

$\Rightarrow \frac{K'_{max}}{k_{max}}$ may not be equal to 2.

Also, $K_{max} = eV_0$ and $K'_{max} = eV'_0$

$\Rightarrow \frac{V'_0}{V_0}$ may not be equal to 2

43. For an ideal gas, PV = nRT

At constant pressure, $P\Delta V = nR\Delta T$

On dividing, $\frac{\Delta V}{V} = \frac{\Delta T}{T}$

$\therefore \delta = \frac{\Delta V}{V\Delta T} = \frac{1}{T}$

Hence graph between δ and T is a rectangular hyperbola.as in option (c).

44. Excess pressures inside the soap bubbles at ends 1 and 2 respectively are

$$p_1 = \frac{2\sigma}{r_1}, p_2 = \frac{2\sigma}{r_2}$$

As $r_1 < r_2$, so, $p_1 > p_2$.

Hence air will flow from end 1 towards end 2.

45. When W = 20 N, $\Delta l = 1 \times 10^{-4}$ m

$$Y = \frac{F}{A}\cdot\frac{l}{\Delta l} = \frac{W}{A}\cdot\frac{l}{\Delta l}$$

$$= \frac{20}{10^{-6}} \times \frac{1}{1\times 10^{-4}}$$

$= 2\times 10^{11} Nm^{-2}$.

46. $t_m = \frac{u\sin\theta}{g} = \frac{1000\times\sin 30^\circ}{9.8} s \simeq 51s.$

47. Magnetic field at the centre of the square loop due to circular loop is given by

$$B = \frac{\mu_0 I R^2}{2[R^2 + (\sqrt{3}R)^2]^{3/2}}$$

$$= \frac{\mu_0 I R^2}{2\times(2^2R^2)^{3/2}} = \frac{\mu_0 I}{16R}$$

Magnetic flux linked with the square loop,

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$$\phi = NBA\cos 45^o = 2 \times \frac{\mu_0 I}{16R} x a^2 \times \frac{1}{\sqrt{2}} = \frac{\mu_0 I a^2}{2^{7/2} R}$$

But $M = \frac{\phi}{I}$ $\therefore \frac{\mu_0 a^2}{2^{p/2} R} = \frac{\mu_0 a^2}{2^{7/2} R}$ $\Rightarrow$ p=7.

48. Applying Ampere's circuital law, we get

$$B_{inside} \times L = \mu_0 I \quad \text{or} B_{in} = \frac{\mu_0 I}{L}$$

$$\varepsilon_{ind} = -\frac{d\phi}{dt} = -\frac{d}{dt}(BA)$$

$$= -\frac{\mu_0}{L}.\pi r^2 \frac{dI}{dt} = -\frac{\mu_0}{L}.\pi r^2 \times 300 I_0 \sin(300t)$$

Magnetic moment,

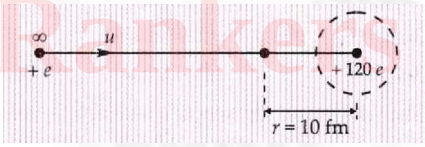
$$|m| = I_{ind}.\pi r^2 = \frac{\varepsilon_{ind}}{R}.\pi r^2$$

$$= \frac{\mu_0.\pi r^2 \times 300 I_0 sin(300t)}{RL}.\pi r^2 = \frac{300\pi^2 r^4}{RL}.\mu_0 I_0 \sin(300t)$$

Given $|m| = N\mu_0 I_0 \sin(300t)$

On comparing, $N = \frac{300\pi^2 r^4}{RL} = \frac{300 \times 9.87 \times (0.1)^2}{0.05 \times 10} \simeq 6.$

49.



At the distance of closest approach,

Loss in K.E. of proton = Gain in P.E.

$$\frac{1}{2} m_p u^2 = \frac{1}{4\pi\varepsilon_0} \frac{(120e)(e)}{r}$$

$$\Rightarrow \frac{p^2}{2m_\rho} = \frac{1}{4\pi\varepsilon_0} \frac{120e^2}{r} \quad \lambda^2 = \frac{h^2}{p^2} = \left(\frac{h}{e}\right)^2 \times 4\pi\varepsilon_0 \times \frac{r}{240 m_p}$$

$$= (4.2 \times 10^{-15})^2 \times \frac{1}{9 \times 10^9} \times \frac{10 \times 10^{-15} \times 3}{240 \times 5 \times 10^{-27}}$$

$$\lambda = 4.2 \times 10^{-15} \times \frac{100}{3 \times 20} m = 7 \times 10^{-15} m = 7fm.$$

50. The experiment measures only the change in length l of the wire.

$Y = \frac{F}{A}.\frac{L}{l}$ The maximum percentage error in Young's modulus of the wire is

$$\frac{\Delta Y}{Y} \times 100 = \frac{\Delta l}{l} \times 100 = \frac{LC}{(45-20)LC} \times 100 = \frac{100}{25} = 4$$

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CHEMISTRY

$1.187 g CaCO_3 + 1.023 g MgCO_3$

51.

$$\underset{xg}{CaCO_3} \xrightarrow{\Delta} CaO + CO_2$$

Moles of $CaCO_3$ decomposed = moles of CaO formed

$\therefore \frac{x}{100}$ = moles of CaO formed

Weight of CaO formed $= \frac{x}{100} \times 56$ ....(1)

{CaO = 40 + 16 = 56}

$MgCO_3 \xrightarrow{\Delta} MgO + CO_2$

$(2.21 - x)\, gm$

Moles of $MgCO_3$ decomposed = moles of MgO

Formed $\frac{(2.21-x)}{84}$ = moles of Mg O formed

Weight of MgO $= \frac{2.21-x}{84} \times 40$ .....(2)

{MgO =24 +16+=40}

Add equation 1 and 2.

$\therefore \frac{x}{100} \times 56 + \frac{(2.21-x)}{84} \times 40 = 1.152$

52.

$2p_x$ *and* $2p_z$ orbitals are degenerate i.e., they are of equal energies and hence no transition of electron between hem is possible

53. CONCEPTUAL

54.

Using the relation, charge $= \frac{Dipole\ moment}{bond\ length} = \frac{0.38 \times 10^{-18}\ esu - cm}{1.61 \times 10^{-8}\ cm} = 2.44 \times 10^{-11}\ esu$

$\therefore percenage\ of\ charge = \frac{2.4 \times 10^{-11}\ esu}{4.8 \times 10^{-10}\ esu} \times 100$

$= 5\%$

55. Both assertion and reason are true and reason is correct explanation of assertion.

$(mol.wt\ of \left[Fe(CN)^{4-}_{6}\right] = 56 + 6 \times 26 = 212$

$Eq.wt = \frac{212}{61} or 3.47$

The balanced equation involves 61 electrons.

$\left[Fe(CN)_6\right]^4 + 30H_2O \rightarrow Fe^{3+} + 6CO_2 + 6NO_3^- + 60H^+ + 61e^-$

56. A – H, S decolourises acidified solution of $KMnO_4$

$2KMnO_4 + 3H_2SO_4 + 5H_2S \rightarrow K_2SO_4 + 2MnSO_4 + 8H_2O + 5S$

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The shape of the H, S molecule is similar to that of the water molecule i.e. V-shaped structure with bond length (H-S) 1.35 A and bond angle (H – S-H) 92.5°involving sp^3 hybridization of S atom

$B \rightarrow SO_2$ undergoes disproportionate (oxidation of two molecules of the same compound at the expense of the third which is reduced) reaction.

$$3SO_2 \xrightarrow{Heat} 2SO_3 + S$$

Decolorises $KMnO_4$

$$5SO_2 + 2KMnO_4 + 2H_2O \rightarrow K_2SO_4 + 2MnSO_4 + 2H_2SO_4$$

Sulphur doxide acts as bleaching agent in the following way :

In the presence of water, it is oxidised with the liberation of nascent hydrogen which reduces the colouring matter to colourless

$$SO_2 + 2H_2O \rightarrow H_2SO_4 + 2[H]$$

Colouring matter + $2[H] \xrightarrow[air]{}$ Colourless compound

Sulphur dioxide has a planar triangular (V –shaped) structure involving sp^2 hybridization of the sulphur atom (sp^2 hybridization occurs between $3s^2 3p_x^1 \, and \, 3p_y^1$ electrons).

$C \rightarrow NO_2$ undergoes decolourisation of $KMnO_4$

$$2KMnO_4 + 3H_2SO_4 + 10NO_2 + 2H_2O \rightarrow K_2SO_4 + 2MnSO_4 + 10HNO_3$$

NO_2 molecule has angular structure (V- shape) with O –N-O bond angle of about 132° and N- O distance of about1.19A which is intermediate between a single and a double bond.

$D \rightarrow HNO_2$ decolourises $KMnO_4$:

$$2KMnO_4 + 3H_2SO_4 + 5HNO_2 \rightarrow K_2SO_4 + 2MnSO_4 + 3H_2O + 5HNO_3$$

X – ray and raman spectral studies indicate and angular structure for nitrite ion (NO_2) with O – N – O angle of 132° and bond length (N –O) of 1.13A.

57. There is a high charge (+3) on aluminium ion (Al^{3+})

And the ratio, $\frac{charge}{surface\ area}$, is high (i.e., Charge density is high)

so it polarizes the H_2O (ligand) and H^+ ions are removed. Due to this (i.e., polarization of Al^{3-} ion) there occurs a drift of electron density towards Al^{3+} ion and thus the O- H bond weakens i.e., the O –H bond in $\left[Al(H_2O)_6\right]^{3+}$ is weaker than in water.

$$(H_2O)Al^{3+} \leftarrow O\begin{matrix} H \\ \curvearrowright \\ H \end{matrix} \longrightarrow \left[(H_2O)_5 Al(OH)_2\right]^{2+} + H^+$$

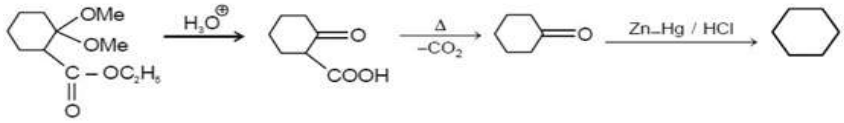
58. CONCEPTUAL

59. $M(AA)_3$ exbhits only optical isomerism

60. $$CHD_2I \xrightarrow{Mg} CHD_2MgI \xrightarrow[D^{\oplus}]{CH_2=O} CHD_2 - CH_2 - OD \xrightarrow[(-HOD)]{Al_2O_3/\Delta} CD_2 - CH_2$$

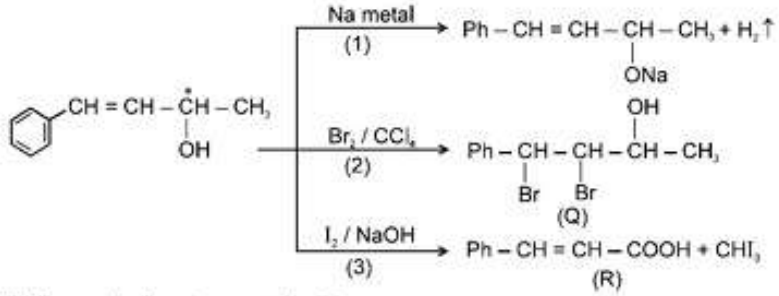
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61.



62. CONCEPTUAL

63.



'P' has optical and geometrical isomers.

64. (III) is most basic as the lone pair of N is localized IV > II > I as per extent of involvement of lone pair of N in resonance. More resonance decreases basic strength due o decreased availability of lone pair for lineis donation.

65. $La^{3+}:[Xe]6S^{\circ}5d^{\circ} \Rightarrow$ No unpaired electrons

$Lu^{3+}:[Xe]6S^{\circ}5d^{\circ} \Rightarrow$ colourless ions

66. Protein gives blue – violet colour with ninhydrin (2, 2 – dihydroxyindane – 1,3 – diene) Carbohydrates gives brown red ppt. With benedict's solution (alkaline $CuSO_4$ citrate

67. (B) –CN is meta directing But –OH is ortho /pera directing.
(E) Halides are deactivating groups.

68. Oxidizing a gent oxidizes HI to I_2

69. The larger the stability, the smaner the P.E,; hence I > II > III.

70. Gabriel phthalimide synthesis –Amine with lesser number of carbon atoms than reactant
Hoffmann bromamide reaction – reaction of phthalimide with with KOH and R- X

71. $E = 1.23 - \frac{0.0591}{4}\log\left[H^+\right]^4 = 1.23 + 0.0591 \times pH$

$= 1.23 + 0.0591 \times 5 = 1.23 + 0.2955 = 1.52V$

72. $A: Fe_{0.93}O \rightarrow Fe_2O_3$

$n - factor = \left(3 - \frac{200}{93}\right) \times 0.93$

n – factor =0.79

B : 2x+(0.93 –x) x 3 = 2

$Fe^{2+} = 0.79, Fe^{3+} = 0.21$

$D: \%Fe^{2+} = \frac{0.79}{0.93} \times 100 = 85\%; Fe^{3+} = 15\%$

73.

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$$\Delta E = R_H Z^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)(n_1 < n_2)$$

$$1.47\times10^{-17} = 2.18\times10^{-18}\times9\left(\frac{1}{n^2} - \frac{1}{(n+1)^2}\right)$$

$$\frac{1.47}{1.96} = \frac{3}{4} = \frac{1}{n^2} - \frac{1}{(n+1)^2}$$

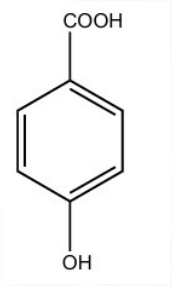
So, n =1

74. Wrok done for reversible isothermal expansion is given as:

$$w = -nRTIn\left(\frac{V_2}{V_1}\right) = -1\times0.08206\times291.15\,In\left(\frac{100}{10}\right)$$ =-55.0.128 L atm

Work done by system $\approx$55 L atm

75.



Molecular weight: 138.

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